

Ch. 12 — P02 (K) Debugging a Flawed Simulation

Problem Bank | Chapter 12 | Type: K | Difficulty:

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Objective

Practice model verification by diagnosing and correcting deliberate bugs in a SimPy simulation. This problem uses a test-driven verification approach: write the tests first, then find the bugs.

The Flawed Model

Below is a SimPy M/M/1 simulation that contains **three deliberate bugs**. Your task is to find and fix all three without being told what they are.

Tasks

1. **Write verification tests first:** Before looking for bugs, write five `pytest` unit tests that a *correct* M/M/1 simulation must pass. At minimum, include:
 - Stability check: when $\rho < 1$, throughput $\approx \lambda$
 - Little's Law: $L \approx \lambda W$
 - Extreme case: when $\lambda \rightarrow 0$, $W_q \rightarrow 0$
 - Formula match: simulated W_q within 5% of $1/(\mu - \lambda) - 1/\mu$
 - Utilization check: $\rho_{\text{sim}} \approx \lambda/\mu$
2. **Run the tests:** Apply your tests to `flawed_mm1` with $\lambda = 3$, $\mu = 4$. Which tests fail? What do the failures suggest about the nature of each bug?
3. **Find and fix all three bugs:** Identify each bug (line number, what it computes incorrectly, what it should compute). Write the corrected function `fixed_mm1`.
4. **Verify the fix:** Re-run all five tests on `fixed_mm1`. All must pass. Include the `pytest` output showing all tests green.

5. **Regression guard:** Add one additional test for each bug you found — a test that would have caught that bug if written before the bug was introduced. Explain why this test targets that specific failure mode.

Submission

Submit a Python file or notebook with:

- The five original verification tests (with docstrings explaining what each tests and why)
- The analysis of which tests failed and what each failure implies
- The corrected `fixed_mm1` function with comments on each fix
- All six regression tests passing